MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project: MansourahMassarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: DESIGN CALCULATION
49PQPOWER DISTRIBUTION: PROCESS PLANT SUBSTN.
FOUNDATION FOR SS - 01

Document Number: MD-216-49PQ-EG-CV-CAL-1626
Revision: D01
Revision Date: 25.10.2023
Status: AS BUILT

Outotec Document Number:	900707	GAU01	WD01	CED01	45007	OU100688912	2
L&T Document Number:	O19045	950	C	K	24310	002	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
C01	15.12.2020	ISSUED FOR CONSTRUCTION	AVG	SJM	AG	
A01	20.10.2020	FIRST ISSUE	AVG	SJM	AG	
D01	25.10.2023	AS BUILT	AVG	SJM	AG	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS……3

1. INTENT……..4

2. EXECUTIVE SUMMARY……4

3. SCOPE……4

4. DESIGN REQUIREMENTS……5

4.1 Design Loads……5

5. CONCRETE FOUNDATIONS……5

5.1 Design……5

5.2 Materials……5

5.3 Foundations for Prefab Substation……5

5.4 Minimum Concrete Cover……5

5.5 Allowable Soil Bearing Capacity……6

6. DESIGN CALCULATION FOR FOUNDATION……..6

References……25

ATTACHMENT1: LOAD DATA DRAWINGS……..25

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This document is to Prepare for design of Foundation for **Prefab Substation (49PQ) SS-01**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Load Data Provided.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Foundation and Civil Design Input attached in ATTACHMENT-1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5 CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Prefab Substation.

Foundation shall be Isolated or Combined Foundation to support the structural framing of prefabricated substation.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Raft & Column	50

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450kN/m^2 at 1.5m depth below EGL (ie. RL 912.25m) refer Table 4 - BH-19 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

6. DESIGN CALCULATION FOR FOUNDATION

As per the loads and layout of the column, four different types of foundation (Isolated and combined) have been designed. For reference, some of the foundation calculations have been attached.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1 of SS#01

Size of Columns	Along-X	Along-Z	ex	ez
Column A	0.70m	0.70m	0.0000m	2.40m
Column B	0.70m	0.70m	0.0000m	2.40m
Size of fdn. raft:	2.00m	7.00m		
Thickness of raft:	0.50 m			
Original Gr. Level:	0.00 m			
Finished Gr. Level:	0.00 m			
U/S of B/Pl Above GL:	3.00 m			
U/S level of fdn:	-1.50 m			
Unit weight of conc:	25 kN/m^3			
Unit weight of soil:	18 kN/m^3			

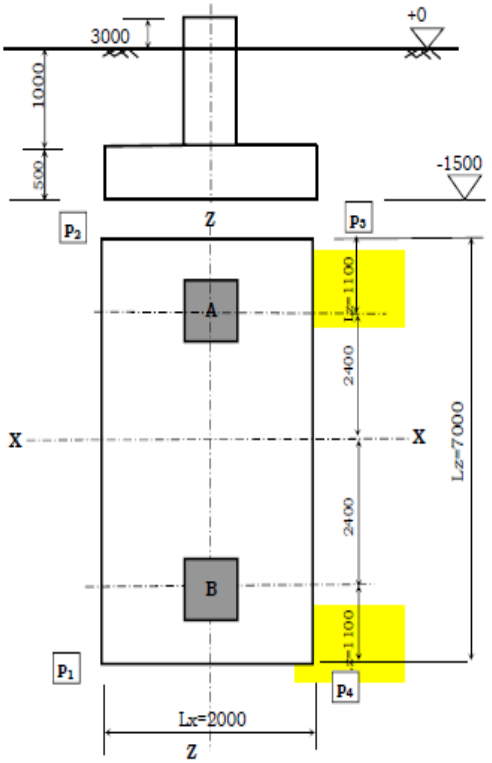
ulation of Weight of Foundation :

Weight of raft = (2*7*0.5*25) = 175.00KN

Weight of soil = ((2*7)-2*(0.7*0.7))*1*18 = 234.36KN

Total = 409.36KN

Submerged wt. = 409.36-2*7*1.5*10 = 199.36KN



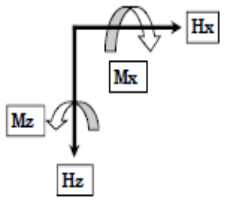
Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 434mm = 0.434m

Effective depth of slab along Z direction = 442mm = 0.442m

Clear projection along X direction = 0.65m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1 of SS#01

Sl. No.	Basic Load Case	Column A (Node N19F) V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	Column B (Node N19E) V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	58.29	3.59	2.79	-8.05	-0.40	194.05	-7.49	12.20	-56.54	-0.77
2	Live Load (LL)	49.81	-3.06	0.84	-18.56	-0.72	184.12	-3.95	2.79	-35.34	-0.46
3	Wind Load (WLx)	-5.01	-1.25	0.01	1.46	0.08	-14.07	1.37	-2.01	0.36	0.01
4	Wind Load (WLz)	-11.46	-2.25	0.05	0.87	0.28	-15.02	1.28	-2.58	-19.07	0.04
5	Earthquake Load (ELx)	0.02	-2.49	-0.16	0.02	0.00	0.09	-0.99	0.07	-0.01	0.00
6	Earthquake Load (ELz)	1.96	-0.13	0.10	0.80	0.10	-0.27	-0.99	0.07	-0.01	0.00
7	Live Load (Roof)(LLr)	12.31	1.60	-0.02	-0.14	0.00	21.85	-2.37	3.63	-1.08	-0.01

Design Loads on each column	Column A					Column B				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 DL+LL	108.09	0.52	3.63	-26.61	-1.12	378.17	-11.44	14.99	-91.87	-1.23
202 DL+0.75LL+0.75LLr	104.87	2.49	3.40	-22.08	-0.94	348.53	-12.23	17.02	-83.85	-1.12
203 DL+WLx	53.27	2.34	2.80	-6.59	-0.31	179.98	-6.12	10.19	-56.18	-0.76
204 DL+WLz	46.83	1.33	2.84	-7.18	-0.11	179.04	-6.21	9.62	-75.60	-0.73
205 DL+0.7ELx	58.30	1.84	2.68	-8.04	-0.40	194.11	-8.18	12.25	-56.54	-0.77
206 DL+0.7ELz	59.66	3.49	2.86	-7.49	-0.33	193.86	-8.18	12.25	-56.54	-0.77
207 DL+0.75WLx+0.75LL+0.75LLr	101.12	1.55	3.41	-20.99	-0.87	337.98	-11.20	15.51	-83.58	-1.11
208 DL+0.75WLz+0.75LL+0.75LLr	96.28	0.80	3.44	-21.43	-0.73	337.27	-11.27	15.08	-98.15	-1.09
209 DL+0.525ELx+0.75LL+0.75LLr	104.88	1.18	3.32	-22.07	-0.94	348.58	-12.75	17.05	-83.86	-1.12
210 DL+0.525ELz+0.75LL+0.75LLr	105.90	2.42	3.45	-21.66	-0.89	348.39	-12.75	17.05	-83.86	-1.12
211 0.6DL+WLx	29.96	0.90	1.69	-3.37	-0.15	102.36	-3.13	5.31	-33.56	-0.45
212 0.6DL+WLz	23.51	-0.10	1.72	-3.96	0.04	101.42	-3.21	4.74	-52.99	-0.42
213 0.6DL+0.7ELx	34.98	0.41	1.56	-4.82	-0.24	116.49	-5.19	7.37	-33.93	-0.46
214 0.6DL+0.7ELz	36.34	2.06	1.74	-4.27	-0.17	116.24	-5.19	7.37	-33.93	-0.46
Ultimate Conditions										
301 1.4DL	81.60	5.02	3.90	-11.27	-0.56	271.67	-10.49	17.08	-79.15	-1.08
302 1.4DL+1.7LL+0.5LLr	172.42	0.61	5.32	-42.90	-1.78	595.60	-18.38	23.64	-139.77	-1.86
303 1.2DL+1.6WLx+LL+0.5LLr	117.89	0.04	4.20	-25.96	-1.06	405.40	-11.93	16.04	-103.15	-1.37
304 1.2DL+1.6WLz+LL+0.5LLr	95.25	-3.17	4.28	-26.76	-0.74	382.03	-9.70	11.48	-133.15	-1.31
305 1.2DL+ELx+LL	119.76	-1.25	4.03	-28.20	-1.20	417.07	-13.92	17.50	-103.19	-1.38
306 1.2DL+ELz+LL	121.71	1.10	4.29	-27.42	-1.10	416.71	-13.92	17.50	-103.19	-1.38
307 0.9DL+1.6WLx	44.44	1.23	2.53	-4.91	-0.22	152.13	-4.55	7.77	-50.31	-0.67
308 0.9DL+1.6WLz	34.12	-0.38	2.59	-5.86	0.09	150.62	-4.69	6.85	-81.39	-0.63
309 0.9DL+ELx	52.47	0.74	2.35	-7.23	-0.36	174.73	-7.73	11.05	-50.89	-0.69
310 0.9DL+ELz	54.42	3.09	2.61	-6.45	-0.26	174.38	-7.73	11.05	-50.89	-0.69

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Design Loads on combined foundation:

L/C	Load at top of foundation					Load at bottom of foundation				
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 DL+LL	486.26	-10.92	18.62	-118.49	-650.54	895.62	-10.92	-30.51	-118.49	-1183.73
202 DL+0.75LL+0.75LLr	453.40	-9.74	20.42	-105.93	-586.83	862.76	-9.74	-23.42	-105.93	-1063.53
203 DL+WLx	233.25	-3.79	13.00	-62.77	-305.17	642.61	-3.79	-4.04	-62.77	-587.61
204 DL+WLz	225.86	-4.88	12.46	-82.78	-318.15	635.22	-4.88	-9.50	-82.78	-690.68
205 DL+0.7ELx	252.41	-6.34	14.93	-64.58	-327.12	661.77	-6.34	-13.61	-64.58	-617.74
206 DL+0.7ELz	253.52	-4.69	15.11	-64.04	-323.18	662.88	-4.69	-6.01	-64.04	-611.34
207 DL+0.75WLx+0.75LL+0.75LLr	439.09	-9.65	18.92	-104.57	-570.46	848.45	-9.65	-24.50	-104.57	-1041.01
208 DL+0.75WLz+0.75LL+0.75LLr	433.55	-10.47	18.52	-119.58	-580.19	842.91	-10.47	-28.60	-119.58	-1118.30
209 DL+0.525ELx+0.75LL+0.75LLr	453.46	-11.57	20.37	-105.93	-586.92	862.82	-11.57	-31.68	-105.93	-1063.60
210 DL+0.525ELz+0.75LL+0.75LLr	454.29	-10.33	20.50	-105.52	-583.97	863.65	-10.33	-25.98	-105.52	-1058.81
211 0.6DL+WLx	132.32	-2.22	7.00	-36.93	-174.37	541.68	-2.22	-3.00	-36.93	-340.56
212 0.6DL+WLz	124.93	-3.32	6.46	-56.95	-187.35	534.29	-3.32	-8.46	-56.95	-443.62
213 0.6DL+0.7ELx	151.48	-4.78	8.93	-38.75	-196.32	560.84	-4.78	-12.57	-38.75	-370.68
214 0.6DL+0.7ELz	152.59	-3.13	9.11	-38.20	-192.38	561.95	-3.13	-4.97	-38.20	-364.29
Ultimate Conditions										
301 1.4DL	353.27	-5.47	20.99	-90.42	-457.80	762.63	-5.47	-3.63	-90.42	-864.70
302 1.4DL+1.7LL+0.5LLr	768.02	-17.77	28.96	-182.67	-1019.26	1177.38	-17.77	-51.01	-182.67	-1841.27
303 1.2DL+1.6WLx+LL+0.5LLr	523.28	-11.89	20.23	-129.10	-692.45	932.64	-11.89	-33.26	-129.10	-1273.43
304 1.2DL+1.6WLz+LL+0.5LLr	477.28	-12.87	15.76	-159.91	-690.31	886.64	-12.87	-42.15	-159.91	-1409.90
305 1.2DL+ELx+LL	536.83	-15.18	21.53	-131.40	-716.11	946.19	-15.18	-46.76	-131.40	-1307.40
306 1.2DL+ELz+LL	538.42	-12.82	21.78	-130.62	-710.48	947.78	-12.82	-35.91	-130.62	-1298.26
307 0.9DL+1.6WLx	196.57	-3.32	10.30	-55.21	-259.37	605.93	-3.32	-4.64	-55.21	-507.83
308 0.9DL+1.6WLz	184.74	-5.07	9.44	-87.24	-280.13	594.10	-5.07	-13.38	-87.24	-672.73
309 0.9DL+ELx	227.21	-6.99	13.40	-58.12	-294.47	636.57	-6.99	-18.07	-58.12	-556.02
310 0.9DL+ELz	228.80	-4.64	13.66	-57.34	-288.85	638.16	-4.64	-7.21	-57.34	-546.88

Check for overturning and sliding:

Load Cases:	Load at B/O Fdn					F.O.S.	F.O.S.	F.O.S.	F.O.S.
	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	Mz	Mx	Hx	Hz
201 DL+LL	895.6	-10.9	-30.5	-118.5	1183.7	29.4	2.6	32.8	3.02
202 DL+0.75LL+0.75LLr	862.8	-9.7	-23.4	-105.9	1063.5	36.8	2.8	35.4	3.26
203 DL+WLx	642.6	-3.8	-4.0	-62.8	587.6	159.2	3.8	67.9	4.10
204 DL+WLz	635.2	-4.9	-9.5	-82.8	690.7	66.9	3.2	52.1	3.07
205 DL+0.7ELx	661.8	-6.3	-13.6	-64.6	617.7	48.6	3.7	41.7	4.10
206 DL+0.7ELz	662.9	-4.7	-6.0	-64.0	611.3	110.3	3.8	56.5	4.14
207 DL+0.75WLx+0.75LL+0.75LL	848.5	-9.6	-24.5	-104.6	1041.0	34.6	2.9	35.2	3.25
208 DL+0.75WLz+0.75LL+0.75LL	842.9	-10.5	-28.6	-119.6	1118.3	29.5	2.6	32.2	2.82
209 DL+0.525ELx+0.75LL+0.75L	862.8	-11.6	-31.7	-105.9	1063.6	27.2	2.8	29.8	3.26
210 DL+0.525ELz+0.75LL+0.75L	863.7	-10.3	-26.0	-105.5	1058.8	33.2	2.9	33.4	3.27
211 0.6DL+WLx	541.7	-2.2	-3.0	-36.9	340.6	180.6	5.6	97.5	5.87
212 0.6DL+WLz	534.3	-3.3	-8.5	-56.9	443.6	63.2	4.2	64.5	3.75
213 0.6DL+0.7ELx	560.8	-4.8	-12.6	-38.7	370.7	44.6	5.3	46.9	5.79
214 0.6DL+0.7ELz	561.9	-3.1	-5.0	-38.2	364.3	113.1	5.4	71.8	5.88
					Min.=	27.2	2.6	29.8	2.8

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD.

Load Combination & Calculation of Base pressure

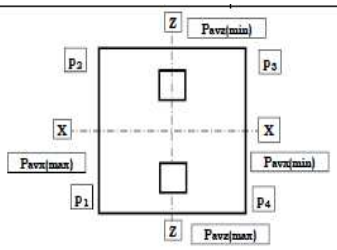
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net safe bearing capacity of soil for DL, LL= 450 kN/m² For Wind load/Seis. load case = 562.5 kN/m²

Gross safe bearing capacity of soil for DL, LL= 477 kN/m² For Wind load/Seis. load case = 589.5 kN/m²



Allowable Stress Design

Load Case	Design Load/moments			Base Pressure (kN/m²)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax+p3)/2	Pavz(max)	Pavz(min)	Pavz(max)	Pavz(min)	Remarks
201 DL+LL	895.62	-30.51	-1183.73	63.97	6.54	72.47	200.00	0.00	200.00	0.00	173.00	100.00	100.00	200.00	0.00	O.K
202 DL+0.75LL+0.75LLr	862.76	-23.42	-1063.53	61.63	5.02	65.11	170.00	0.00	170.00	0.00	143.00	85.00	85.00	170.00	0.00	O.K
203 DL+WLx	642.61	-4.04	-587.61	45.90	0.87	35.98	110.00	0.00	110.00	0.00	83.00	55.00	55.00	110.00	0.00	O.K
204 DL+WLz	635.22	-9.50	-690.68	45.37	2.04	42.29	120.00	0.00	120.00	0.00	93.00	60.00	60.00	120.00	0.00	O.K
205 DL+0.7ELx	661.77	-13.61	-617.74	47.27	2.92	37.82	110.00	0.00	105.00	0.00	83.00	55.00	52.50	107.50	0.00	O.K
206 DL+0.7ELz	662.88	-6.01	-611.34	47.35	1.29	37.43	110.00	0.00	105.00	0.00	83.00	55.00	52.50	107.50	0.00	O.K
207 DL+0.75WLx+0.75LL+0.75LLr	848.45	-24.50	-1041.01	60.60	5.25	63.74	175.00	0.00	175.00	0.00	148.00	87.50	87.50	175.00	0.00	O.K
208 DL+0.75WLz+0.75LL+0.75LLr	842.91	-28.60	-1118.30	60.21	6.13	68.47	175.00	0.00	175.00	0.00	175.00	87.50	87.50	175.00	0.00	O.K
209 DL+0.525ELx+0.75LL+0.75LLr	862.82	-31.68	-1063.60	61.63	6.79	65.12	165.00	0.00	165.00	0.00	138.00	82.50	82.50	165.00	0.00	O.K
210 DL+0.525ELz+0.75LL+0.75LLr	863.65	-25.98	-1058.81	61.69	5.57	64.82	165.00	0.00	165.00	0.00	138.00	82.50	82.50	165.00	0.00	O.K
211 0.6DL+WLx	541.68	-3.00	-340.56	38.69	0.64	20.85	60.18	18.48	58.90	17.20	33.18	39.33	38.05	59.54	17.84	O.K
212 0.6DL+WLz	534.29	-8.46	-443.62	38.16	1.81	27.16	100.00	0.00	100.00	0.00	73.00	50.00	50.00	100.00	0.00	O.K
213 0.6DL+0.7ELx	560.84	-12.57	-370.68	40.06	2.69	22.69	65.45	20.06	60.06	14.67	38.45	42.76	37.37	62.76	17.37	O.K
214 0.6DL+0.7ELz	561.95	-4.97	-364.29	40.14	1.07	22.30	63.51	18.90	61.38	16.77	36.51	41.21	39.08	62.45	17.84	O.K

Ultimate Conditions																
301 1.4DL	762.63	-3.63	-864.70	54.47	0.78	52.94	110.00	0.00	110.00	0.00	83.00	55.00	55.00	110.00	0.00	O.K
302 1.4DL+1.7LL+0.5LLr	1177.38	-51.01	-1841.27	84.10	10.93	112.73	225.00	0.00	225.00	0.00	198.00	112.50	112.50	225.00	0.00	O.K
303 1.2DL+1.6WLx+LL+0.5LLr	932.64	-33.26	-1273.43	66.62	7.13	77.96	160.00	0.00	160.00	0.00	133.00	80.00	80.00	160.00	0.00	O.K
304 1.2DL+1.6WLz+LL+0.5LLr	886.64	-42.15	-1409.90	63.33	9.03	86.32	185.00	0.00	185.00	0.00	158.00	92.50	92.50	185.00	0.00	O.K
305 1.2DL+ELx+LL	946.19	-46.76	-1307.40	67.59	10.02	80.04	170.00	0.00	170.00	0.00	143.00	85.00	85.00	170.00	0.00	O.K
306 1.2DL+ELz+LL	947.78	-35.91	-1298.26	67.70	7.69	79.49	170.00	0.00	170.00	0.00	143.00	85.00	85.00	170.00	0.00	O.K
307 0.9DL+1.6WLx	605.93	-4.64	-507.83	43.28	1.00	31.09	75.37	13.18	73.38	11.19	48.37	44.28	42.29	74.38	12.19	O.K
308 0.9DL+1.6WLz	594.10	-13.38	-672.73	42.44	2.87	41.19	85.00	0.00	85.00	0.00	58.00	42.50	42.50	85.00	0.00	O.K
309 0.9DL+ELx	636.57	-18.07	-556.02	45.47	3.87	34.04	85.00	0.00	85.00	0.00	58.00	42.50	42.50	85.00	0.00	O.K
310 0.9DL+ELz	638.16	-7.21	-546.88	45.58	1.55	33.48	85.00	0.00	85.00	0.00	58.00	42.50	42.50	85.00	0.00	O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1 of SS#01

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.7+2*0.5*0.442)+(0.7+2*0.5*0.442))*2

= 4.568 m

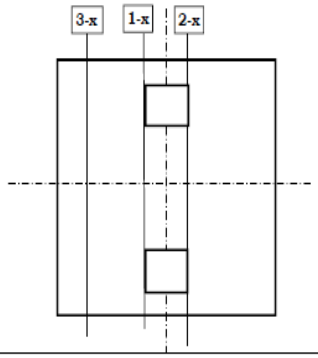
Punching Area= ((0.7+2*0.5*0.442)*(0.7+2*0.5*0.442))

= 1.3 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*25+1*0.5*18)

= 30.50 kN/m^2

Area of footing, A_f= 14.00 m^2



DESIGN OF FOUNDATION MKD. F1 of SS#01

Load Case	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
301 1.4DL	55.00	55.00	55.00	5.18	5.18	5.29	10.10
302 1.4DL+1.7LL+0.5LLr	112.50	112.50	112.50	17.32	17.32	17.71	26.17
303 1.2DL+1.6WLx+LL+0.5LLr	80.00	80.00	80.00	10.46	10.46	10.69	13.89
304 1.2DL+1.6WLz+LL+0.5LLr	92.50	92.50	92.50	13.10	13.10	13.39	-25.00
305 1.2DL+ELx+LL	85.00	85.00	85.00	11.51	11.51	11.77	9.26
306 1.2DL+ELz+LL	85.00	85.00	85.00	11.51	11.51	11.77	11.21
307 0.9DL+1.6WLx	43.63	42.94	44.07	2.87	2.54	2.95	-12.28
308 0.9DL+1.6WLz	42.50	42.50	42.50	2.54	2.54	2.59	-21.13
309 0.9DL+ELx	42.50	42.50	42.50	2.54	2.54	2.59	-2.78
310 0.9DL+ELz	42.50	42.50	42.50	2.54	2.54	2.59	-0.83

Overhang = 0.65 m

Effective depth = 0.434 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	17.32	0.09	0.12	0.000	126.97	781.20	16ϕT	257.38	300	200	1005.31
At top	0.00	0.00	0.00	0.000	0.00	390.60	16ϕT	514.75	300	200	1005.31

DESIGN OF FOUNDATION MKD. F1 of SS#01

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 17.71 kN/m

Shear stress , τ_v= (17.71*1000)/(1000*434)

= 0.04 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 26.17 kN

Strength of Concrete in shear Vc1 = 5864.35 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 3812.00 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 3909.57 kN **Table 22.6.5.2_ (c)**

Design Vc= 3812.00 kN

Allowable shear stress, (ϕ Vc)/bd = 1.44 N/mm^2

Design punching shear stress, τ = (26.17*1000)/(4568*442)

= 0.01 N/mm^2 **No shear reinforcement reqd.**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1 of SS#01

Calculation of distribution reinf in Z direction :

Effective depth = 0.442 m

Location	Mux (kN-m)	ϕ Kn = M_{ui}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	8.00	0.04	0.05	0.000	57.51	795.60	16ϕT	252.72	300	200	1005.31
At top	23.00	0.12	0.16	0.000	165.63	397.80	16ϕT	505.43	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 25.50 kN/m

Shear stress , τ_v= (25.50*1000)/(1000*442)

= 0.06 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	200	c/c
At Top	16ϕT	200	c/c
Along Z Direction :			
At Bottom	16ϕT	200	c/c
At Top	16ϕT	200	c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F2 of SS#01

Size of Columns	Along-X	Along-Z	ex	ez
Column A	0.70m	0.70m	0.0000m	4.05m
Column B	0.70m	0.70m	0.0000m	0.70m
Column C	0.70m	0.70m	0.0000m	4.05m
Size of fdn. raft:	2.50m	9.50m		
Thickness of raft:	0.50 m			
Original Gr. Level:	0.00 m			
Finished Gr. Level:	0.00 m			
U/S of B/Pl Above GL:	3.00 m			
U/S level of fdn:	-1.50 m			
Unit weight of conc:	25 kN/m³			
Unit weight of soil:	18 kN/m³			

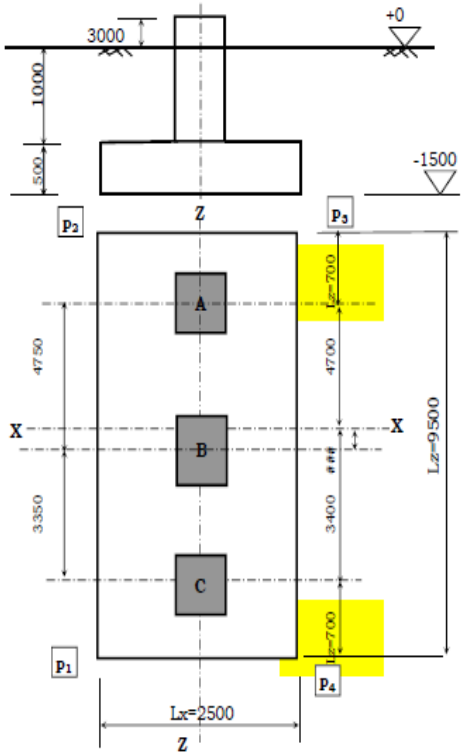
Calculation of Weight of Foundation :

Weight of raft = (2.5*9.5*0.5*25) = 296.88KN

Weight of soil = ((2.5*9.5)-2*(0.7*0.7))*1*18 = 409.86KN

Total = 706.74KN

Submerged wt. = 706.74-2.5*9.5*1.5*10 = 350.49KN



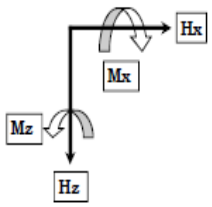
Design Input:

Characteristic strength of concrete, f'c=	35 MPa
Yield stress of steel reinforcement, fy=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 434mm = 0.434m

Effective depth of slab along Z direction = 442mm = 0.442m

Clear projection along X direction = 0.9m



H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F2 of SS#01

Sl. No.	Basic Load Case	Column A (Node N19F) V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	Column B (Node N19E) V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	Column C (Node N04A) V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	58.29	3.59	2.79	-8.05	-0.40	194.05	-7.49	12.20	-56.54	-0.77	100.23	-4.14	2.81	62.73	1.70
2	Live Load (LL)	49.81	-3.06	0.84	-18.56	-0.72	184.12	-3.95	2.79	-35.34	-0.46	68.35	-7.96	0.00	51.37	1.37
3	Wind Load (WLx)	-5.01	-1.25	0.01	1.46	0.08	-14.07	1.37	-2.01	0.36	0.01	-6.99	-1.67	-0.10	-1.90	-0.06
4	Wind Load (WLz)	-11.46	-2.25	0.05	0.87	0.28	-15.02	1.28	-2.58	-19.07	0.04	-10.90	-1.90	-0.28	1.02	0.39
5	Earthquake Load (ELx)	0.02	-2.49	-0.16	0.02	0.00	0.09	-0.99	0.07	-0.01	0.00	0.03	-2.51	-0.14	-0.02	0.00
6	Earthquake Load (ELz)	1.96	-0.13	0.10	0.80	0.10	-0.27	-0.99	0.07	-0.01	0.00	-1.68	0.08	-0.11	1.00	0.11
7	Live Load (Roof)(LLr)	12.31	1.60	-0.02	-0.14	0.00	21.85	-2.37	3.63	-1.08	-0.01	15.23	2.07	0.19	0.84	-0.05

Design Loads on each column	Column A					Column B					Column C				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design															
201 DL+LL	108.09	0.52	3.63	-26.61	-1.12	378.17	-11.44	14.99	-91.87	-1.23	168.58	-12.10	2.81	114.10	3.07
202 DL+0.75LL+0.75LLr	104.87	2.49	3.40	-22.08	-0.94	348.53	-12.23	17.02	-83.85	-1.12	162.91	-8.56	2.95	101.89	2.69
203 DL+WLx	53.27	2.34	2.80	-6.59	-0.31	179.98	-6.12	10.19	-56.18	-0.76	93.24	-5.82	2.71	60.83	1.64
204 DL+WLz	46.83	1.33	2.84	-7.18	-0.11	179.04	-6.21	9.62	-75.60	-0.73	89.33	-6.04	2.53	63.75	2.09
205 DL+0.7ELx	58.30	1.84	2.68	-8.04	-0.40	194.11	-8.18	12.25	-56.54	-0.77	100.25	-5.90	2.71	62.71	1.70
206 DL+0.7ELz	59.66	3.49	2.86	-7.49	-0.33	193.86	-8.18	12.25	-56.54	-0.77	99.05	-4.09	2.73	63.43	1.77
207 DL+0.75WLx+0.75LL+0.75LLr	101.12	1.55	3.41	-20.99	-0.87	337.98	-11.20	15.51	-83.58	-1.11	157.67	-9.81	2.88	100.46	2.65
208 DL+0.75WLz+0.75LL+0.75LLr	96.28	0.80	3.44	-21.43	-0.73	337.27	-11.27	15.08	-98.15	-1.09	154.74	-9.98	2.74	102.65	2.98
209 DL+0.525ELx+0.75LL+0.75LLr	104.88	1.18	3.32	-22.07	-0.94	348.58	-12.75	17.05	-83.86	-1.12	162.93	-9.88	2.88	101.87	2.69
210 DL+0.525ELz+0.75LL+0.75LLr	105.90	2.42	3.45	-21.66	-0.89	348.39	-12.75	17.05	-83.86	-1.12	162.03	-8.52	2.90	102.41	2.75
211 0.6DL+WLx	29.96	0.90	1.69	-3.37	-0.15	102.36	-3.13	5.31	-33.56	-0.45	53.15	-4.16	1.59	35.74	0.96
212 0.6DL+WLz	23.51	-0.10	1.72	-3.96	0.04	101.42	-3.21	4.74	-52.99	-0.42	49.24	-4.38	1.41	38.66	1.41
213 0.6DL+0.7ELx	34.98	0.41	1.56	-4.82	-0.24	116.49	-5.19	7.37	-33.93	-0.46	60.16	-4.25	1.59	37.62	1.02
214 0.6DL+0.7ELz	36.34	2.06	1.74	-4.27	-0.17	116.24	-5.19	7.37	-33.93	-0.46	58.96	-2.43	1.61	38.34	1.10
Ultimate Conditions															
301 1.4DL	81.60	5.02	3.90	-11.27	-0.56	271.67	-10.49	17.08	-79.15	-1.08	140.32	-5.80	3.94	87.82	2.38
302 1.4DL+1.7LL+0.5LLr	172.42	0.61	5.32	-42.90	-1.78	595.60	-18.38	23.64	-139.77	-1.86	264.13	-18.29	4.03	175.57	4.68
303 1.2DL+1.6WLx+LL+0.5LLr	117.89	0.04	4.20	-25.96	-1.06	405.40	-11.93	16.04	-103.15	-1.37	185.06	-14.57	3.31	124.02	3.29
304 1.2DL+1.6WLz+LL+0.5LLr	95.25	-3.17	4.28	-26.76	-0.74	382.03	-9.70	11.48	-133.15	-1.31	163.57	-17.00	2.83	127.86	4.05
305 1.2DL+ELx+LL	119.76	-1.25	4.03	-28.20	-1.20	417.07	-13.92	17.50	-103.19	-1.38	188.66	-15.44	3.24	126.62	3.40
306 1.2DL+ELz+LL	121.71	1.10	4.29	-27.42	-1.10	416.71	-13.92	17.50	-103.19	-1.38	186.94	-12.85	3.26	127.65	3.52
307 0.9DL+1.6WLx	44.44	1.23	2.53	-4.91	-0.22	152.13	-4.55	7.77	-50.31	-0.67	79.02	-6.41	2.37	53.41	1.44
308 0.9DL+1.6WLz	34.12	-0.38	2.59	-5.86	0.09	150.62	-4.69	6.85	-81.39	-0.63	72.77	-6.76	2.08	58.09	2.15
309 0.9DL+ELx	52.47	0.74	2.35	-7.23	-0.36	174.73	-7.73	11.05	-50.89	-0.69	90.24	-6.24	2.39	56.43	1.53
310 0.9DL+ELz	54.42	3.09	2.61	-6.45	-0.26	174.38	-7.73	11.05	-50.89	-0.69	88.52	-3.65	2.42	57.46	1.64

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Design Loads on combined foundation:

L/C	Load at top of foundation V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Load at bottom of foundation V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 DL+LL	654.84	-23.02	21.43	-4.39	-1093.10	1361.58	-23.02	-82.15	-4.39	-1112.85
202 DL+0.75LL+0.75LLr	616.32	-18.30	23.37	-4.05	-986.17	1323.06	-18.30	-58.97	-4.05	-1004.39
203 DL+WLx	326.49	-9.60	15.71	-1.94	-512.59	1033.23	-9.60	-27.49	-1.94	-521.32
204 DL+WLz	315.19	-10.92	14.99	-19.04	-534.21	1021.93	-10.92	-34.14	-19.04	-619.87
205 DL+0.7ELx	352.66	-12.24	17.64	-1.87	-549.52	1059.40	-12.24	-37.45	-1.87	-557.95
206 DL+0.7ELz	352.57	-8.78	17.84	-0.61	-542.85	1059.31	-8.78	-21.68	-0.61	-545.58
207 DL+0.75WLx+0.75LL+0.75LLr	596.76	-19.46	21.80	-4.11	-958.63	1303.50	-19.46	-65.77	-4.11	-977.11
208 DL+0.75WLz+0.75LL+0.75LLr	588.29	-20.45	21.26	-16.93	-974.84	1295.03	-20.45	-70.75	-16.93	-1051.02
209 DL+0.525ELx+0.75LL+0.75LLr	616.39	-21.44	23.25	-4.06	-986.33	1323.13	-21.44	-73.24	-4.06	-1004.58
210 DL+0.525ELz+0.75LL+0.75LLr	616.32	-18.85	23.40	-3.11	-981.32	1323.06	-18.85	-61.41	-3.11	-995.31
211 0.6DL+WLx	185.47	-6.38	8.59	-1.19	-292.87	892.21	-6.38	-20.12	-1.19	-298.24
212 0.6DL+WLz	174.16	-7.70	7.87	-18.29	-314.48	880.90	-7.70	-26.77	-18.29	-396.80
213 0.6DL+0.7ELx	211.64	-9.02	10.52	-1.13	-329.79	918.38	-9.02	-30.08	-1.13	-334.87
214 0.6DL+0.7ELz	211.55	-5.56	10.72	0.14	-323.12	918.29	-5.56	-14.31	0.14	-322.51
Ultimate Conditions										
301 1.4DL	493.59	-11.27	24.93	-2.61	-769.04	1200.33	-11.27	-25.79	-2.61	-780.77
302 1.4DL+1.7LL+0.5LLr	1032.16	-36.06	32.99	-7.10	-1712.82	1738.90	-36.06	-129.29	-7.10	-1744.76
303 1.2DL+1.6WLx+LL+0.5LLr	708.34	-26.46	23.54	-5.08	-1163.55	1415.08	-26.46	-95.51	-5.08	-1186.41
304 1.2DL+1.6WLz+LL+0.5LLr	640.85	-29.87	18.59	-32.05	-1159.44	1347.59	-29.87	-115.80	-32.05	-1303.66
305 1.2DL+ELx+LL	725.49	-30.62	24.76	-4.78	-1203.26	1432.23	-30.62	-113.02	-4.78	-1224.76
306 1.2DL+ELz+LL	725.36	-25.67	25.05	-2.97	-1193.72	1432.10	-25.67	-90.48	-2.97	-1207.09
307 0.9DL+1.6WLx	275.59	-9.73	12.68	-1.80	-435.63	982.33	-9.73	-31.09	-1.80	-443.73
308 0.9DL+1.6WLz	257.51	-11.83	11.52	-29.16	-470.21	964.25	-11.83	-41.72	-29.16	-601.41
309 0.9DL+ELx	317.45	-13.24	15.79	-1.69	-494.68	1024.19	-13.24	-43.77	-1.69	-502.29
310 0.9DL+ELz	317.32	-8.29	16.08	0.12	-485.14	1024.06	-8.29	-21.23	0.12	-484.62

Check for overturning and sliding:

Load Cases:	Load at B/O Fdn V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL+LL	1361.6	-23.0	-82.1	-4.4	1112.9	20.7	5.8	23.7	124.09
202 DL+0.75LL+0.75LLr	1323.1	-18.3	-59.0	-4.0	1004.4	28.0	6.3	28.9	130.75
203 DL+WLx	1033.2	-9.6	-27.5	-1.9	521.3	47.0	9.4	43.0	213.13
204 DL+WLz	1021.9	-10.9	-34.1	-19.0	619.9	37.4	7.8	37.4	21.47
205 DL+0.7ELx	1059.4	-12.2	-37.5	-1.9	557.9	35.4	9.0	34.6	226.24
206 DL+0.7ELz	1059.3	-8.8	-21.7	-0.6	545.6	61.1	9.2	48.3	697.33
207 DL+0.75WLx+0.75LL+0.75L	1303.5	-19.5	-65.8	-4.1	977.1	24.8	6.3	26.8	126.98
208 DL+0.75WLz+0.75LL+0.75L	1295.0	-20.4	-70.8	-16.9	1051.0	22.9	5.9	25.3	30.60
209 DL+0.525ELx+0.75LL+0.75L	1323.1	-21.4	-73.2	-4.1	1004.6	22.6	6.3	24.7	130.47
210 DL+0.525ELz+0.75LL+0.75L	1323.1	-18.8	-61.4	-3.1	995.3	26.9	6.3	28.1	170.31
211 0.6DL+WLx	892.2	-6.4	-20.1	-1.2	298.2	55.4	14.2	55.9	298.72
212 0.6DL+WLz	880.9	-7.7	-26.8	-18.3	396.8	41.1	10.5	45.8	19.26
213 0.6DL+0.7ELx	918.4	-9.0	-30.1	-1.1	334.9	38.2	13.0	40.7	325.48
214 0.6DL+0.7ELz	918.3	-5.6	-14.3	0.1	322.5	80.2	13.5	66.0	2685.01
					Min.=	20.7	5.8	23.7	19.3

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD.

Load Combination & Calculation of Base pressure

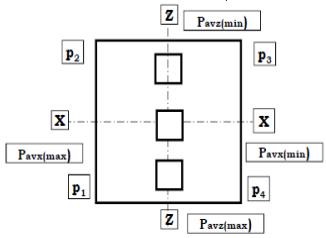
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net safe bearing capacity of soil for DL, LL= 450 kN/m² For Wind load/Seis. load case = 562.5 kN/m²

Gross safe bearing capacity of soil for DL, LL= 477 kN/m² For Wind load/Seis. load case = 589.5 kN/m²



Allowable Stress Design

Load Case	Design Load/moments			Base Pressure (kN/m²)												Remarks
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	
201 DL+LL	1361.58	-82.15	-1112.85	57.33	8.30	29.59	95.22	36.04	78.62	19.43	68.22	65.63	49.03	86.92	27.74	O.K
202 DL+0.75LL+0.75LLr	1323.06	-58.97	-1004.39	55.71	5.96	26.71	88.38	34.96	76.46	23.04	61.38	61.67	49.75	82.42	29.00	O.K
203 DL+WLx	1033.23	-27.49	-521.32	43.50	2.78	13.86	60.15	32.42	54.59	26.86	33.15	46.29	40.73	57.37	29.64	O.K
204 DL+WLz	1021.93	-34.14	-619.87	43.03	3.45	16.48	62.96	29.99	56.06	23.09	35.96	46.48	39.58	59.51	26.54	O.K
205 DL+0.7ELx	1059.40	-37.45	-557.95	44.61	3.78	14.84	63.23	33.55	55.66	25.98	36.23	48.39	40.82	59.45	29.77	O.K
206 DL+0.7ELz	1059.31	-21.68	-545.58	44.60	2.19	14.51	61.30	32.28	56.92	27.90	34.30	46.79	42.41	59.11	30.09	O.K
207 DL+0.75WLx+0.75LL+0.75LLr	1303.50	-65.77	-977.11	54.88	6.65	25.98	87.51	35.55	74.22	22.25	60.51	61.53	48.24	80.87	28.90	O.K
208 DL+0.75WLz+0.75LL+0.75LLr	1295.03	-70.75	-1051.02	54.53	7.15	27.95	89.63	33.73	75.33	19.43	175.00	61.68	47.38	82.48	26.58	O.K
209 DL+0.525ELx+0.75LL+0.75LLr	1323.13	-73.24	-1004.58	55.71	7.40	26.71	89.83	36.40	75.02	21.59	62.83	63.12	48.31	82.43	29.00	O.K
210 DL+0.525ELz+0.75LL+0.75LLr	1323.06	-61.41	-995.31	55.71	6.21	26.47	88.38	35.45	75.97	23.03	61.38	61.92	49.50	82.18	29.24	O.K
211 0.6DL+WLx	892.21	-20.12	-298.24	37.57	2.03	7.93	47.53	31.67	43.46	27.60	20.53	39.60	35.53	45.50	29.64	O.K
212 0.6DL+WLz	880.90	-26.77	-396.80	37.09	2.70	10.55	50.35	29.24	44.94	23.83	23.35	39.80	34.39	47.65	26.54	O.K
213 0.6DL+0.7ELx	918.38	-30.08	-334.87	38.67	3.04	8.91	50.61	32.80	44.53	26.72	23.61	41.71	35.63	47.57	29.76	O.K
214 0.6DL+0.7ELz	918.29	-14.31	-322.51	38.66	1.45	8.58	48.69	31.53	45.80	28.64	21.69	40.11	37.22	47.25	30.09	O.K
Ultimate Conditions																
301 1.4DL	1200.33	-25.79	-780.77	50.54	2.61	20.76	73.91	32.38	68.70	27.17	46.91	53.15	47.94	71.31	29.78	O.K
302 1.4DL+1.7LL+0.5LLr	1738.90	-129.29	-1744.76	73.22	13.07	46.40	132.68	39.88	106.55	13.75	105.68	86.28	60.15	119.62	26.82	O.K
303 1.2DL+1.6WLx+LL+0.5LLr	1415.08	-95.51	-1186.41	59.58	9.65	31.55	100.78	37.68	81.48	18.38	73.78	69.23	49.93	91.13	28.03	O.K
304 1.2DL+1.6WLz+LL+0.5LLr	1347.59	-115.80	-1303.66	56.74	11.70	34.67	103.11	33.77	79.71	10.37	76.11	68.44	45.04	91.41	22.07	O.K
305 1.2DL+ELx+LL	1432.23	-113.02	-1224.76	60.30	11.42	32.57	104.29	39.16	81.45	16.31	77.29	71.73	48.88	92.87	27.74	O.K
306 1.2DL+ELz+LL	1432.10	-90.48	-1207.09	60.30	9.14	32.10	101.54	37.34	83.26	19.06	74.54	69.44	51.16	92.40	28.20	O.K
307 0.9DL+1.6WLx	982.33	-31.09	-443.73	41.36	3.14	11.80	56.30	32.70	50.02	26.42	29.30	44.50	38.22	53.16	29.56	O.K
308 0.9DL+1.6WLz	964.25	-41.72	-601.41	40.60	4.22	15.99	60.81	28.82	52.38	20.39	33.81	44.82	36.39	56.60	24.61	O.K
309 0.9DL+ELx	1024.19	-43.77	-502.29	43.12	4.42	13.36	60.90	34.19	52.06	25.34	33.90	47.55	38.70	56.48	29.77	O.K
310 0.9DL+ELz	1024.06	-21.23	-484.62	43.12	2.15	12.89	58.15	32.38	53.86	28.09	31.15	45.27	40.98	56.01	30.24	O.K

DESIGN OF FOUNDATION MKD. F2 of SS#01

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m².

p_{2x} =pressure at the face of column at section 2x-2x in kN/m².

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m².

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.7+2*0.5*0.442)+(0.7+2*0.5*0.442))*2

= 4.568 m

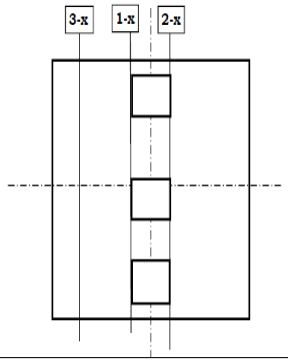
Punching Area= ((0.7+2*0.5*0.442)*(0.7+2*0.5*0.442))

= 1.3 m² Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*25+1*0.5*18)

= 30.50 kN/m²

Area of footing, A_f= 23.75 m²



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F2 of SS#01

Load Case	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
301 1.4DL	51.27	49.82	52.18	8.92	7.32	10.33	14.94
302 1.4DL+1.7LL+0.5LLr	76.87	69.56	81.41	21.32	13.28	24.86	72.49
303 1.2DL+1.6WLx+LL+0.5LLr	62.28	56.88	65.63	14.75	8.81	17.21	36.92
304 1.2DL+1.6WLz+LL+0.5LLr	60.02	53.46	64.08	14.23	7.03	16.66	17.23
305 1.2DL+ELx+LL	63.50	57.11	67.47	15.59	8.55	18.22	37.21
306 1.2DL+ELz+LL	62.86	57.74	66.03	14.88	9.26	17.35	39.99
307 0.9DL+1.6WLx	42.24	40.48	43.33	5.36	3.43	6.25	-10.47
308 0.9DL+1.6WLz	41.79	39.42	43.25	5.39	2.80	6.31	-20.20
309 0.9DL+ELx	44.36	41.89	45.90	6.48	3.75	7.56	-5.20
310 0.9DL+ELz	43.73	42.52	44.47	5.77	4.45	6.70	-2.42

Overhang = 0.9 m
Effective depth = 0.434 m

Location	Mux (kN-m)	ϕ Kn = M_u/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	21.32	0.11	0.15	0.000	156.36	781.20	16ϕT	257.38	300	200	1005.31
At top	0.00	0.00	0.00	0.000	0.00	390.60	16ϕT	514.75	300	200	1005.31

DESIGN OF FOUNDATION MKD. F2 of SS#01

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 24.86 kN/m

Shear stress , τ_v= (24.86*1000)/(1000*434)

= 0.06 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 72.49 kN

Strength of Concrete in shear Vc1 = 5864.35 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 3812.00 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 3909.57 kN **Table 22.6.5.2_ (c)**

Design Vc= 3812.00 kN

Allowable shear stress, (ϕ Vc)/bd = 1.44 N/mm^2

Design punching shear stress, τ = (72.49*1000)/(4568*442)

= 0.04 N/mm^2 **No shear reinforcement reqd.**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F2 of SS#01

Calculation of distribution reinf in Z direction :

Effective depth = 0.442 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	6.40	0.03	0.04	0.000	46.00	795.60	16ϕT	252.72	300	200	1005.31
At top	18.40	0.09	0.13	0.000	132.44	397.80	16ϕT	505.43	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 20.40 kN/m

Shear stress , τ_v= (20.40*1000)/(1000*442)

= 0.05 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	200	c/c
At Top	16ϕT	200	c/c
Along Z Direction :			
At Bottom	16ϕT	200	c/c
At Top	16ϕT	200	c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F3 of SS#01

	Along-X	Along-Z
Size of column :	0.70m	0.70m
Size of fdn. raft :	3.30m	3.50m
Thickness of raft :	0.50 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Wind Load (X)	WLx
Wind Load (Z)	WLz
Earthquake Load (X)	ELx
Earthquake Load (Z)	ELz
Live Load (Roof)	LLr

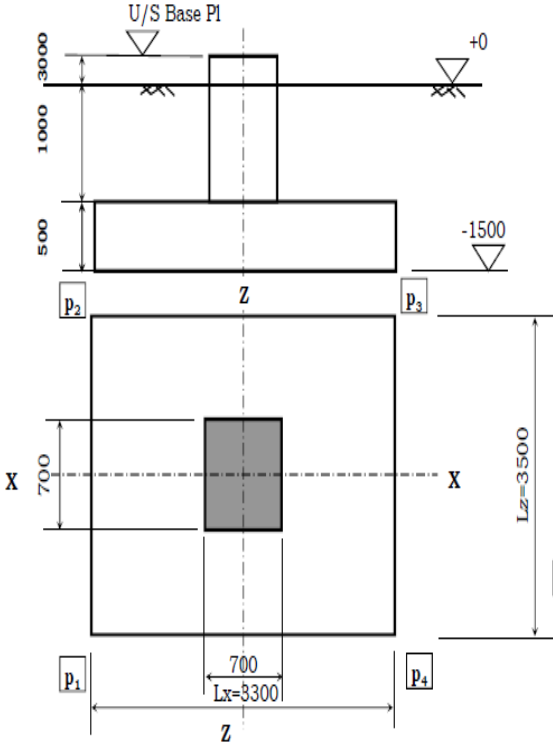
Calculation of Weight of Foundation :

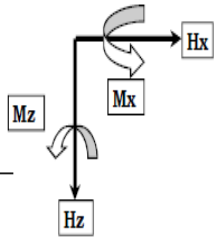
Weight of raft & Col.= (3.3*3.5*0.5*24)+(0.7*0.7*4*24) = 185.64KN
Weight of soil = ((3.3*3.5)-(0.7*0.7))*1*18 = 199.08KN
Total = 384.72KN

Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	25mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	25mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 437.5mm = 0.4375m
Effective depth of slab along Z direction = 425mm = 0.425m
Clear projection along Z direction = 1.4m
Clear projection along X direction = 1.3m





H_x=Force along X-X direction
Hz=Force along Z-Z direction
M_x=Moment about X-X direction
Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F3 of SS#01 (Critical Node N16F)

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	56.76	2.36	-4.43	-35.88	-0.68
2	Live Load (LL)	66.48	8.39	-1.78	-81.92	-1.58
3	Wind Load (WLx)	-5.09	-1.24	-0.36	8.07	0.21
4	Wind Load (WLz)	-3.93	-1.64	1.34	18.90	0.59
5	Earthquake Load (ELx)	0.28	-3.18	-1.83	0.05	0.00
6	Earthquake Load (ELz)	4.64	0.10	-0.03	1.97	0.14
7	Live Load (Roof)(LLr)	11.78	1.08	-0.57	-13.35	-0.18

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+LL	123.24	10.75	-6.21	-117.80	-2.26	507.96	10.75	42.15	-117.80	-532.36
202	DL+0.75LL+0.75LLr	115.45	9.47	-6.19	-107.33	-2.00	500.17	9.47	36.40	-107.33	-485.00
203	DL+WLx	51.66	1.13	-4.79	-27.81	-0.47	436.38	1.13	0.28	-27.81	-125.62
204	DL+WLz	52.83	0.72	-3.10	-16.99	-0.09	437.55	0.72	0.16	-16.99	-76.52
205	DL+0.7ELx	56.95	0.14	-5.71	-35.85	-0.68	441.67	0.14	-5.09	-35.85	-162.00
206	DL+0.7ELz	60.01	2.43	-4.45	-34.50	-0.58	444.73	2.43	6.49	-34.50	-155.83
207	DL+0.75WLx+0.75LL+0.75LLr	111.63	8.54	-6.46	-101.28	-1.84	496.35	8.54	31.96	-101.28	-457.61
208	DL+0.75WLz+0.75LL+0.75LLr	112.50	8.24	-5.19	-93.16	-1.55	497.22	8.24	31.87	-93.16	-420.78
209	DL+0.525ELx+0.75LL+0.75LLr	115.59	7.80	-7.15	-107.31	-2.00	500.31	7.80	27.93	-107.31	-484.88
210	DL+0.525ELz+0.75LL+0.75LLr	117.89	9.52	-6.21	-106.30	-1.92	502.61	9.52	36.62	-106.30	-480.26
211	0.6DL + WLx	28.96	0.18	-3.02	-13.46	-0.20	413.68	0.18	-2.20	-13.46	-60.77
212	0.6DL + WLz	30.13	-0.22	-1.32	-2.63	0.18	414.85	-0.22	-2.32	-2.63	-11.67
213	0.6DL + 0.7ELx	34.25	-0.81	-3.94	-21.50	-0.40	418.97	-0.81	-7.57	-21.50	-97.14
214	0.6DL + 0.7ELx	37.31	1.49	-2.68	-20.15	-0.31	422.03	1.49	4.01	-20.15	-90.97
Ultimate Conditions											
301	1.4 DL	79.46	3.31	-6.21	-50.23	-0.95	464.18	3.31	8.67	-50.23	-227.00
302	1.4 DL + 1.7 LL +0.5LLr	198.36	18.11	-9.52	-196.17	-3.73	583.08	18.11	71.96	-196.17	-886.50
303	1.2 DL+1.6WLx+LL +0.5LLr	132.33	9.79	-7.95	-118.74	-2.15	517.05	9.79	36.09	-118.74	-536.49
304	1.2 DL+1.6WLz+LL +0.5LLr	134.19	9.14	-5.25	-101.42	-1.54	518.91	9.14	35.89	-101.42	-457.93
305	1.2 DL+ELx+LL	134.86	8.04	-8.93	-124.93	-2.39	519.58	8.04	27.27	-124.93	-564.58
306	1.2 DL+ELz+LL	139.23	11.32	-7.13	-123.00	-2.26	523.95	11.32	43.81	-123.00	-555.77
307	0.9DL+ 1.6WLx	42.93	0.15	-4.56	-19.38	-0.27	427.65	0.15	-3.89	-19.38	-87.50
308	0.9DL+ 1.6WLz	44.79	-0.50	-1.85	-2.06	0.34	429.51	-0.50	-4.09	-2.06	-8.94
309	0.9DL+ ELx	51.36	-1.05	-5.82	-32.25	-0.61	436.08	-1.05	-10.55	-32.25	-145.72
310	0.9DL+ Elz	55.73	2.23	-4.02	-30.32	-0.47	440.45	2.23	5.99	-30.32	-136.91

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Check for overturning and sliding:

Load Cases:		Load at B/O Fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	DL+LL	507.96	10.75	42.15	-117.80	-532.36	19.88	1.57	18.90	1.72
202	DL+0.75LL+0.75LLr	500.17	9.47	36.40	-107.33	-485.00	22.67	1.70	21.14	1.86
203	DL+WLx	436.38	1.13	0.28	-27.81	-125.62	2567.88	5.73	154.91	6.28
204	DL+WLz	437.55	0.72	0.16	-16.99	-76.52	4639.81	9.43	242.14	10.30
205	DL+0.7ELx	441.67	0.14	-5.09	-35.85	-162.00	143.06	4.50	1282.81	4.93
206	DL+0.7ELz	444.73	2.43	6.49	-34.50	-155.83	113.10	4.71	73.15	5.16
207	DL+0.75WLx+0.75LL+0.75LLr	496.35	8.54	31.96	-101.28	-457.61	25.62	1.79	23.25	1.96
208	DL+0.75WLz+0.75LL+0.75LLr	497.22	8.24	31.87	-93.16	-420.78	25.74	1.95	24.15	2.13
209	DL+0.525ELx+0.75LL+0.75LLr	500.31	7.80	27.93	-107.31	-484.88	29.55	1.70	25.67	1.86
210	DL+0.525ELz+0.75LL+0.75LLr	502.61	9.52	36.62	-106.30	-480.26	22.65	1.73	21.12	1.89
211	0.6DL + WLx	413.68	0.18	-2.20	-13.46	-60.77	310.60	11.23	908.99	12.29
212	0.6DL + WLz	414.85	-0.22	-2.32	-2.63	-11.67	294.73	58.67	747.60	63.01
213	0.6DL + 0.7ELx	418.97	-0.81	-7.57	-21.50	-97.14	91.30	7.12	207.66	7.80
214	0.6DL + 0.7ELx	422.03	1.49	4.01	-20.15	-90.97	173.64	7.65	113.52	8.38

DESIGN OF FOUNDATION MKD. F3 of SS#01

Load Combination & Calculation of Base pressure

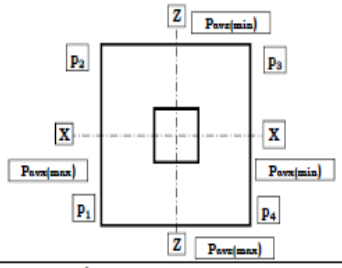
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 477 kN/m² For WL/SL case = 589.5 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)					
201	DL+LL	507.96	42.15	532.36	43.98	6.64	79.02	185.00	0.00	185.00	0.00	158.00	92.50	92.50	185.00	0.00	O.K
202	DL+0.75LL+0.75LLr	500.17	36.40	485.00	43.30	5.73	71.98	160.00	0.00	160.00	0.00	133.00	80.00	80.00	160.00	0.00	O.K
203	DL+WLx	436.38	0.28	125.62	37.78	0.04	18.65	56.47	19.18	56.38	19.09	29.47	37.83	37.74	56.43	19.14	O.K
204	DL+WLz	437.55	0.16	76.52	37.88	0.02	11.36	49.27	26.55	49.22	26.50	22.27	37.91	37.86	49.25	26.53	O.K
205	DL+0.7ELx	441.67	5.09	162.00	38.24	0.80	24.04	63.09	15.00	61.48	13.39	36.09	39.05	37.44	62.29	14.20	O.K
206	DL+0.7ELz	444.73	6.49	155.83	38.50	1.02	23.13	62.65	16.40	60.61	14.35	35.65	39.53	37.48	61.63	15.38	O.K
207	DL+0.75WLx+0.75LL+0.75LLr	496.35	31.96	457.61	42.97	5.03	67.92	145.00	0.00	145.00	0.00	118.00	72.50	72.50	145.00	0.00	O.K
208	DL+0.75WLz+0.75LL+0.75LLr	497.22	31.87	420.78	43.05	5.02	62.45	135.00	0.00	135.00	0.00	108.00	67.50	67.50	135.00	0.00	O.K
209	DL+0.525ELx+0.75LL+0.75LLr	500.31	27.93	484.88	43.32	4.40	71.97	165.00	0.00	165.00	0.00	138.00	82.50	82.50	165.00	0.00	O.K
210	DL+0.525ELz+0.75LL+0.75LLr	502.61	36.62	480.26	43.52	5.76	71.28	160.00	0.00	160.00	0.00	133.00	80.00	80.00	160.00	0.00	O.K
211	0.6DL + WLx	413.68	2.20	60.77	35.82	0.35	9.02	45.18	27.14	44.49	26.45	18.18	36.16	35.47	44.84	26.80	O.K
212	0.6DL + WLz	414.85	2.32	11.67	35.92	0.37	1.73	38.01	34.55	37.28	33.82	11.01	36.28	35.55	37.65	34.19	O.K
213	0.6DL + 0.7ELx	418.97	7.57	97.14	36.27	1.19	14.42	51.88	23.05	49.50	20.66	24.88	37.47	35.08	50.69	21.86	O.K
214	0.6DL + 0.7ELx	422.03	4.01	90.97	36.54	0.63	13.50	50.67	23.67	49.41	22.41	23.67	37.17	35.91	50.04	23.04	O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

301	1.4 DL	464.18	8.67	227.00	40.19	1.37	33.69	75.25	7.86	72.52	5.13	48.25	41.56	38.83	73.89	6.50	O.K
302	1.4 DL + 1.7 LL +0.5LLr	583.08	71.96	886.50	50.48	11.33	131.58	520.00	0.00	520.00	0.00	493.00	260.00	260.00	520.00	0.00	O.K
303	1.2 DL+1.6WLx+LL +0.5LLr	517.05	36.09	536.49	44.77	5.68	79.63	150.00	0.00	150.00	0.00	123.00	75.00	75.00	150.00	0.00	O.K
304	1.2 DL+1.6WLz+LL +0.5LLr	518.91	35.89	457.93	44.93	5.65	67.97	130.00	0.00	130.00	0.00	103.00	65.00	65.00	130.00	0.00	O.K
305	1.2 DL+ELx+LL	519.58	27.27	564.58	44.99	4.29	83.80	170.00	0.00	170.00	0.00	143.00	85.00	85.00	170.00	0.00	O.K
306	1.2 DL+ELz+LL	523.95	43.81	555.77	45.36	6.90	82.49	170.00	0.00	170.00	0.00	143.00	85.00	85.00	170.00	0.00	O.K
307	0.9DL+ 1.6WLx	427.65	3.89	87.50	37.03	0.61	12.99	50.62	24.65	49.40	23.43	23.62	37.64	36.42	50.01	24.04	O.K
308	0.9DL+ 1.6WLz	429.51	4.09	8.94	37.19	0.64	1.33	39.16	36.50	37.87	35.22	12.16	37.83	36.55	38.52	35.86	O.K
309	0.9DL+ ELx	436.08	10.55	145.72	37.76	1.66	21.63	61.04	17.79	57.72	14.47	34.04	39.42	36.10	59.38	16.13	O.K
310	0.9DL+ Elz	440.45	5.99	136.91	38.13	0.94	20.32	59.40	18.76	57.51	16.87	32.40	39.08	37.19	58.46	17.82	O.K
	Max :	508	42	532													
	Min :	436	0	77													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 158.00 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 168.00 kN/m²
Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 133.00 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 143.00 kN/m²

Sample calculation of values of base pressure:

Load case : DL+0.75LL+0.75LLr

Maximum pressure, p_1 (pmax) = 43.30+5.73+71.98= 121.02 kN/m²

p_2= 43.30+5.73-71.98= -22.95 kN/m²

p_3= 43.30-(5.73+71.98)= -34.41 kN/m²

p_4= 43.30-5.73+71.98= 109.56 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 160.00-1.5*18= 133.00 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (160.00+0.00)/2= 49.03 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+160.00)/2= 37.57 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (160.00+160.00)/2= 115.29 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+0.00)/2= -28.68 kN/m²

DESIGN OF FOUNDATION MKD. F3 of SS#01

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m².

p_{2x} =pressure at the face of column at section 2x-2x in kN/m².

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m².

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

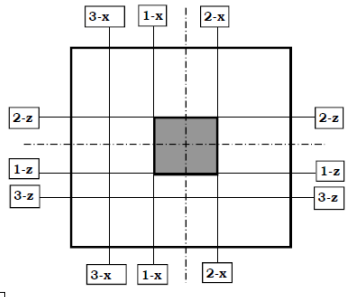
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.7+2*0.5*0.425)+(0.7+2*0.5*0.425))*2
= 4.5 m

Punching Area= ((0.7+2*0.5*0.425)*(0.7+2*0.5*0.425))
= 1.27 m² Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*24+1*18)
= 30.00 kN/m²

Area of footing, A_f= 11.55 m²



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
	Ultimate Conditions							
301	1.4 DL	40.48	39.91	40.85	9.47	7.76	9.66	104.74
302	1.4 DL + 1.7 LL +0.5LLr	260.00	260.00	260.00	194.35	194.35	198.38	210.57
303	1.2 DL+1.6WLx+LL +0.5LLr	75.00	75.00	75.00	38.03	38.03	38.81	151.79
304	1.2 DL+1.6WLz+LL +0.5LLr	65.00	65.00	65.00	29.58	29.58	30.19	153.45
305	1.2 DL+ELx+LL	85.00	85.00	85.00	46.48	46.48	47.44	154.05
306	1.2 DL+ELz+LL	85.00	85.00	85.00	46.48	46.48	47.44	157.94
307	0.9DL+ 1.6WLx	37.16	36.90	37.32	6.32	5.56	6.45	72.23
308	0.9DL+ 1.6WLz	37.33	37.05	37.50	6.47	5.68	6.61	73.89
309	0.9DL+ ELx	38.11	37.41	38.55	7.59	5.52	7.75	79.73
310	0.9DL+ Elz	38.34	37.93	38.59	7.46	6.29	7.62	83.62

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 301

p_{1x} = 39.53-(39.53-37.48)*1.3/3.3 = 38.72 kN/m^2 Overhang = 1.3 m

p_{2x} = 37.48+(39.53-37.48)*1.3/3.3 = 38.29 kN/m^2 Effective depth = 0.4375 m

p_{3x} = 39.53-(39.53-37.48)*(1.3-0.4375)/3.3 = 38.99 kN/m^2

BM_{1x} = (40.48-30)*1.3*1.3/2+0.5*(39.53-40.48)*1.3*1.3*0.67 = 8.32 kN-m/m

BM_{2x} = (39.91-30)*1.3*1.3/2-0.5*(-37.48+39.91)*1.3*1.3*0.67 = 7.00 kN-m/m

SF_{3x} = (40.85-30)*(1.3-0.4375)+0.5*(39.53-40.85)*(1.3-0.4375) = 8.79 kN/m

$S.F_{(punching)}$ = (11.55-1.27)*(38.50-30) = 87.43 kN

Location	Mux (kN-m)	ϕ Kn = M_{ui}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	194.35	1.02	1.35	0.003	1443.89	787.50	25ϕT	339.97	300	125	3926.99
At top	0.00	0.00	0.00	0.000	0.00	393.75	16ϕT	510.63	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 198.38 kN/m

Shear stress , τ_v= (198.38*1000)/(1000*437.5)

= 0.45 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 210.57 kN

Strength of Concrete in shear Vc1 = 5823.64 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 3828.50 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 3882.43 kN **Table 22.6.5.2_ (c)**

Design Vc= 3828.50 kN

Allowable shear stress, (ϕ Vc)/bd = 1.46 N/mm^2

Design punching shear stress, τ = (210.57*1000)/(4500*437.5)

= 0.11 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
	Ultimate Conditions						
301	1.4 DL	46.93	33.46	55.12	34.29	-14.31	33.64
302	1.4 DL + 1.7 LL +0.5LLr	312.00	208.00	375.14	412.93	37.87	407.13
303	1.2 DL+1.6WLx+LL +0.5LLr	90.00	60.00	108.21	98.20	-10.00	96.63
304	1.2 DL+1.6WLz+LL +0.5LLr	78.00	52.00	93.79	81.18	-12.58	79.85
305	1.2 DL+ELx+LL	102.00	68.00	122.64	115.21	-7.41	113.41
306	1.2 DL+ELz+LL	102.00	68.00	122.64	115.21	-7.41	113.41
307	0.9DL+ 1.6WLx	39.62	34.43	42.78	16.25	-2.48	15.98
308	0.9DL+ 1.6WLz	37.46	36.92	37.78	8.01	6.09	7.95
309	0.9DL+ ELx	42.08	33.43	47.33	23.20	-8.00	22.77
310	0.9DL+ Elz	42.20	34.08	47.14	22.63	-6.68	22.23

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 301

p_{1z} = 61.63-(61.63-15.38)*1.4/3.5= 43.13 kN/m^2

p_{2z} = 15.38+(61.63-15.38)*1.4/3.5= 33.88 kN/m^2

p_{3z} = 61.63-(61.63-15.38)*(1.4-0.425)/3.5= 48.75 kN/m^2

Overhang = 1.4 m

Effective depth = 0.425 m

BM_{1z} = (46.93-30)*1.4*1.4/2+0.5*(61.63-46.93)*1.4*1.4*0.67 = 26.24 kN-m/m

BM_{2z} = (33.46-30)*1.4*1.4/2-0.5*(-15.38+33.46)*1.4*1.4*0.67 = -8.48 kN-m/m

SF_{3z} = (55.12-30)*(1.4-0.425)+0.5*(61.63-55.12)*(1.4-0.425) = 27.66 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	412.93	2.286	3.048	0.0077	3357.01	765.00	25ϕT	146.22	300	125	3927.0
At top	-14.31	-0.079	-0.106	-0.0003	-1.07	382.50	16ϕT	525.65	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 407.13 kN/m

Shear stress , τ_v= (407.13*1000)/(1000*425)

= 0.96 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	25ϕT	125	c/c
	At Top	16ϕT	200	c/c
Along Z Direction :				
	At Bottom	25ϕT	125	c/c
	At Top	16ϕT	200	c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

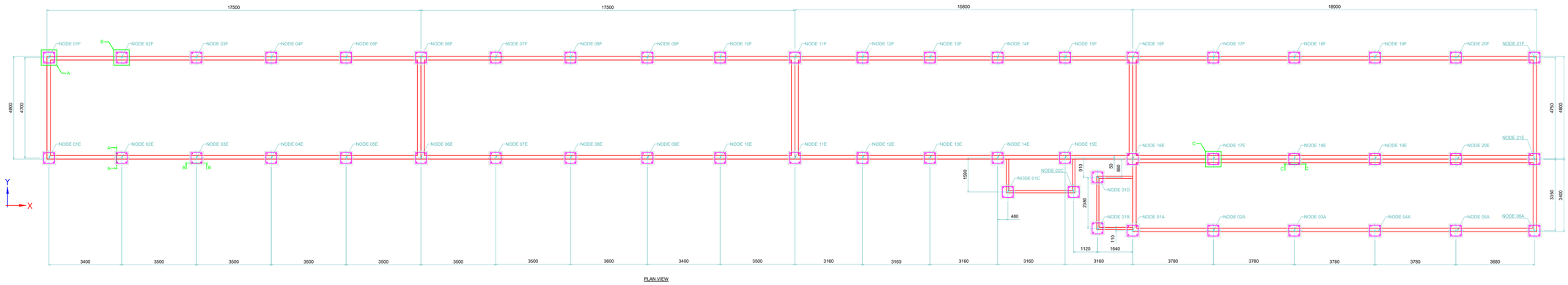
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

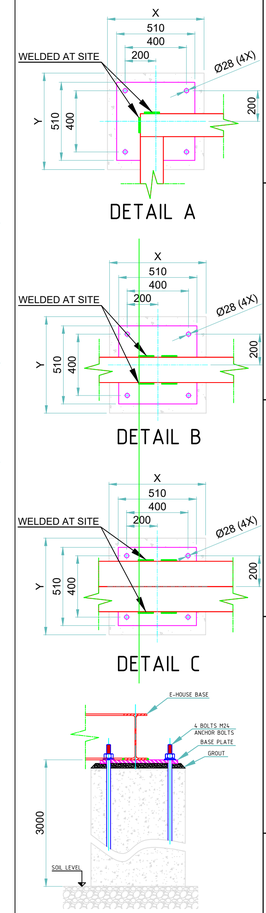
ATTACHMENT1: LOAD DATA DRAWINGS

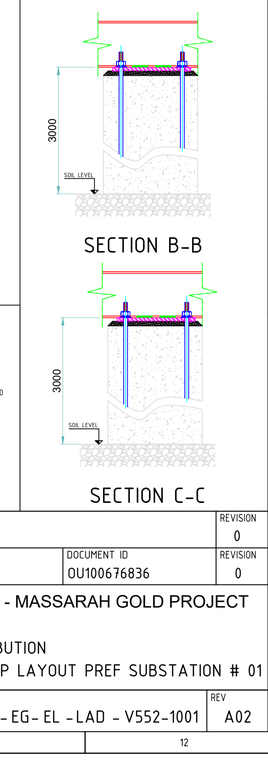
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of OutotecOyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DRAWING NUMBER	REV
MD-216-49PQ-EG-EL-LAD-V552-1001	A00







GENERAL NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE
2. LOAD PARAMETERS FOR WIND AND SEISMIC CONDITIONS TAKEN FROM DOCUMENT MD-216-0000-EG-ST-CRT-0001
3. NUMBER OF PILLARS: 52
4. DIMENSIONS "X" AND "Y" SHOWED IN THE DETAILS "A" AND "B" SHALL BE DEFINED ACCORDING TO CONCRETE AND BUILD CHARACTERISTICS
5. CONCRETE FOUNDATIONS ARE NOT PART OF SIEMENS SCOPE
6. THE BOLTS SHOWED IN THE SECTIONS "A-A" AND "B-B" SHALL BE CAST INTO THE CONCRETE
7. ANCHOR BOLTS SHALL BE FABRICATED AS SPECIFIED IN DOCUMENT MD-101-SMEM-EG-CV-SPC-002
8. THE E-HOUSE BASE SHALL BE WELDED TO THE BASE PLATE ON SITE
9. THE CONCRETE COLUMNS SHALL BE FINISHED WITH 30 mm GROUT UNDER STEEL COLUMN BASE PLATE

1.0 Dead Load

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 1-A	2938	5056	31843	198.8	-9028.4	267.5
Node 2-A	2420	40777	55084	867.6	-1699.2	1674.0
Node 3-A	2489	47382	81660	1332.7	-5095.2	1226.2
Node 4-A	4143	62726	100230	1696.8	2813.6	-1216.3
Node 5-A	1624	70808	75986	1294.8	12715.0	-1742.2
Node 6-A	4415	7319	33354	14376.0	9172.7	875.3
Node 1-B	735	1123	12108	1018.8	-346.8	-899.9
Node 1-C	1577	-8672	10768	-231.2	-679.8	-663.6
Node 2-C	3240	-8086	12205	442.5	862.0	1290.0
Node 1-D	27382	-4845	12224	-923.5	229.2	356.0
Node 1-E	4798	9085	43326	14725.0	-9992.5	-537.7
Node 2-E	2852	29927	61136	6931.0	-2662.3	-591.8
Node 3-E	10614	40009	69615	775.3	-282.7	1344.1
Node 4-E	5342	41803	78989	703.8	-5697.8	-622.0
Node 5-E	-2943	38504	84702	648.8	647.4	569.6
Node 6-E	-90	44378	81261	13182.0	-1422.6	-12.3
Node 7-E	2438	36499	84236	737.0	535.0	975.1
Node 8-E	6324	39164	77256	781.9	3392.2	1254.7
Node 9-E	-56	31802	74183	523.4	4836.8	-835.4
Node 10-E	5060	29580	65646	680.5	1418.9	577.0
Node 11-E	866	49135	43981	385.0	2948.1	1317.1
Node 12-E	1408	38507	40128	790.8	-22.2	1647.0
Node 13-E	-3337	35392	39263	751.8	303.3	1663.5
Node 14-E	518	36223	57466	659.7	-6936.2	1006.3
Node 15-E	-9452	20127	71014	291.0	1325.5	2337.5
Node 16-E	12669	34713	96512	3305.8	-1388.9	717.5
Node 17-E	24974	-2899	144430	-118.5	-602.0	-347.0
Node 18-E	-5992	-43716	161180	-592.4	-3278.2	-197.2
Node 19-E	7493	-56535	194050	-768.7	5201.5	241.1
Node 20-E	28784	-59982	143290	-599.7	18140.0	1903.7
Node 21-E	-798	-9380	46162	-6461.7	3037.4	-575.9
Node 1-F	6678	-8765	49411	-17298.0	-3094.0	540.3
Node 2-F	2925	-29648	54645	-690.5	-3968.8	151.1
Node 3-F	10065	-40378	84689	-812.3	-3355.1	-1418.6
Node 4-F	5770	-41220	114180	-845.6	-13593.0	-429.4
Node 5-F	-3013	-41846	129650	-790.8	6813.3	-122.3
Node 6-F	28	-42349	91241	-13550.0	2471.7	53.9
Node 7-F	-1925	-36493	88066	-747.5	-2875.3	982.4
Node 8-F	6278	-39596	86522	-799.2	1402.3	-1364.4
Node 9-F	364	-31472	81282	-531.2	5184.3	766.9
Node 10-F	4746	-29599	67035	-694.3	1115.2	-577.6
Node 11-F	1053	-49029	44630	-432.2	1948.8	-1333.1
Node 12-F	1210	-37923	41203	-804.3	313.2	-1602.3
Node 13-F	-4094	-32649	40102	-769.2	-513.6	-1535.1
Node 14-F	-4919	-27917	48069	-779.8	-1782.5	-974.7
Node 15-F	-2239	-20131	49096	-700.2	-2146.5	-594.1
Node 16-F	2362	-35881	56158	-678.1	-4433.5	982.4
Node 17-F	-3026	-26747	56455	-575.5	2894.2	-848.5
Node 18-F	-3221	-9383	53505	-390.4	-500.7	-139.7
Node 19-F	3085	-8052	58285	-396.4	2789.0	-234.6
Node 20-F	6533	-7981	45875	-258.5	1697.2	589.1
Node 21-F	2692	1548	22755	-6446.5	7787.8	-90.0

1.0 Live Load

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 1-A	-3932	11992	40676	454.7	-9421.0	437.8
Node 2-A	1047	63930	68379	1439.2	-2382.6	2704.4
Node 3-A	9131	45807	68072	1329.8	578.2	87.2
Node 4-A	-7957	51373	68350	1368.5	1381.7	-866.7
Node 5-A	-9776	83876	69290	1537.2	3622.2	-876.3
Node 6-A	1982	826	35791	18061.0	12959.0	849.9
Node 1-B	-25	-797	6901	1984.2	-1003.1	-1851.2
Node 1-C	-6164	-22438	-4701	-946.8	-1054.9	-1633.7
Node 2-C	11039	-21816	-5514	-1425.7	1384.1	2780.3
Node 1-D	40300	-10325	5883	-1776.1	-213.3	400.0
Node 1-E	4062	14191	82588	32509.0	-24054.0	-1201.3
Node 2-E	13434	82811	106660	1967.2	-5270.4	-909.8
Node 3-E	27431	94178	163110	1804.8	1554.4	2954.7
Node 4-E	10634	89037	163950	1692.9	-6638.2	-1466.0
Node 5-E	-6237	76035	153730	1434.7	-358.6	1098.8
Node 6-E	389	101450	167930	33041.0	1.4	-202.1
Node 7-E	-4442	81092	156897	1570.4	-781.2	-2162.1
Node 8-E	14035	90953	157900	1629.9	3099.0	3127.2
Node 9-E	2793	84610	164920	1376.7	8929.9	-2195.6
Node 10-E	9941	69711	118890	1675.6	1213.0	1227.6
Node 11-E	2057	110650	77732	2149.3	3388.2	3157.7
Node 12-E	-496	96387	72620	2041.7	-127.2	3866.0
Node 13-E	-4677	89634	72370	1950.8	479.2	3920.0
Node 14-E	2095	80572	95679	1563.2	-12149.0	2283.6
Node 15-E	-27725	47220	116240	660.8	13229.0	5850.9
Node 16-E	16597	83329	164650	9201.5	-6429.4	2309.1
Node 17-E	-61247	-6756	192790	-112.4	-1220.1	-713.4
Node 18-E	-7951	-35578	180540	-453.2	-2598.8	562.2
Node 19-E	-3945	-50339	184120	-457.4	5950.0	-118.7
Node 20-E	-14919	-43468	185790	-446.0	3426.3	304.8
Node 21-E	-28977	-4142	68875	-7753.5	14843.0	-42.0
Node 1-F	-4243	-13780	83154	-32050.0	-24159.0	-1149.9
Node 2-F	13233	-83193	156340	-1958.6	-5659.0	879.5
Node 3-F	27355	-94457	163160	-1803.7	1479.9	-2951.0
Node 4-F	10667	-89728	164020	1702.7	-7009.5	1429.5
Node 5-F	-5570	-76554	153880	-1410.1	-647.4	-1203.6
Node 6-F	352	-100760	167700	-31562.0	-547.0	322.8
Node 7-F	-4460	-81788	156900	-1728.7	579.3	2160.6
Node 8-F	13934	-91016	157820	-1918.3	2619.4	-3158.9
Node 9-F	3157	-84159	164960	-1322.8	8334.3	2107.1
Node 10-F	9941	-69711	118890	-1675.6	1213.0	-1227.6
Node 11-F	2122	-110040	77540	-2199.4	3015.7	-3472.5
Node 12-F	-1144	-94539	73322	-2050.9	-511.6	-3727.0
Node 13-F	-7284	-82458	71814	-1999.5	1104.4	-3623.5
Node 14-F	-12431	-61799	68849	-1894.4	133.1	-2201.6
Node 15-F	-7259	-45658	65907	-1768.5	1002.5	-1736.0
Node 16-F	-8387	-81920	164277	-1581.8	-1791.2	2492.6
Node 17-F	-174	-36123	53047	-857.7	2219.4	-1219.7
Node 18-F	-7949	-19092	49258	-674.4	-554.2	312.6
Node 19-F	-3065	-18562	49805	-720.9	839.0	-177.1
Node 20-F	610	-38703	52212	-860.3	1490.1	443.1
Node 21-F	3391	4291	27445	-10595.0	10837.0	-195.4

1.0 Wind Load Transversal

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 1-A	-3710	-1461	-11891	158.0	1304.3	1376.8
Node 2-A	-626	736	-9919	282.8	169.6	16.3
Node 3-A	-1176	139	-10821	324.8	-466.9	-681.7
Node 4-A	-1895	1022	-10901	389.2	-280.9	362.7
Node 5-A	-3055	4224	-10096	373.9	-467.0	702.5
Node 6-A	-1446	-10325	-9123	5825.3	-3183.4	-697.5
Node 1-B	1414	-5090	-5707	2329.5	672.6	-474.4
Node 1-C	-3784	-16274	-7081	3452.4	1670.2	-834.9
Node 2-C	5568	-14467	-5126	3994.6	-809.2	439.4
Node 1-D	5776	-5388	3947	2356.5	54.1	-108.4
Node 1-E	-3107	-29437	-37465	6297.6	8970.2	-700.2
Node 2-E	2219	-11194	-31982	420.5	-22.4	893.0
Node 3-E	2687	-22549	-26725	9343.7	1112.4	-364.3
Node 4-E	-1295	-30191	-51914	438.3	10050.0	846.7
Node 5-E	-2437	-30450	9558.3	9300.8	556.5	-223.1
Node 6-E	-37073	-30990	5834.2	1112.8	-255.8	
Node 7-E	-23919	-35115	9817.1	-1693.3	754.0	
Node 8-E	-12504	-30550	950.6	-1536.7	-990.6	
Node 9-E	-41910	-55365	101689.0	-2663.7	1920.9	
Node 10-E	2099	4736	-28789.8	2238.7	-106.9	
Node 11-E	-20257	-15549	4915.9	-315.1	-304.7	
Node 12-E	-6006	-16338	420.4	11.3	-1639.3	
Node 13-E	-19136	-37896	10267.0	-3094.0	236.0	
Node 14-E	-7190	-10929	208.9	-2733.2	-521.2	
Node 15-E	-10130	34.9	311.4	93.9		
Node 16-E	-15010	-10228	-2060.4	-1365.2	380.4	
Node 17-E	-23474	-17378	-17.8	-1281.5	-1022.3	
Node 18-E	-17755	-12494	43.1	-1162.1	-130.3	
Node 19-E	-19306	-15015	40.6	-2502.8	127.9	
Node 20-E	-35656	-13850	-15.7	-2899.9	303.5	
Node 21-E	-18814	-8425	-3883.1	-238.2	230.5	
Node 1-F	-28171	-41332	7985.0	-4157.6	-722.1	
Node 2-F	-31100	30441	408.8	278.0	864.9	
Node 3-F	-23035	27068	8963.7	-954.5	404.4	
Node 4-F	-32626	32503	350.3	-4424.2	904.7	
Node 5-F	-24471	29080	9213.8	675.0	-539.4	
Node 6-F	-31379	5424.3	-1214.9	-317.1		
Node 7-F	-23910	34346	9487.9	1795.3	154.3	
Node 8-F	-33383	500.6	-193.3	-1167.5		
Node 9-F	-52899	9883.2	4115.6	1733.7		
Node 10-F	-18237	29883	10402.0	2178.9		
Node 11-F	-41332	3773.3	-390.6	-366.5		
Node 12-F	-6726	308.2	-656.5	-1400.3		
Node 13-F	-8180	-7345	270.7	-374.1	-1146.6	
Node 14-F	-6606	-4045	-7566	235.1	-56.5	-255.8
Node 15-F	374	1359	-6608	279.9	28.9	262.5
Node 16-F	-5539	18896	-3930	5911.9	1136.5	555.6
Node 17-F	-6671	-10591	254.3	320.9	29.2	
Node 18-F	388	433	-11315	253.2	89.6	-502.3
Node 19-F	-1254	869	-11460	282.3	49.9	254.5
Node 20-F	-2255	3812	-9859	257.5	-1404.4	256.0
Node 21-F	-2911	-5983	-337	8533.9	-2358.6	-656.0

1.0 Wind Load Longitudinal

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 1-A	-2029	-896	-3604	-57.6	160.8	-282.6
Node 2-A	-1331	-1790	-6124	-59.2	615.0	-34.5
Node 3-A	-1612	-1755	-6974	-52.3	-83.8	60.2
Node 4-A	-1673	-1899	-6991	-55.4	-99.6	-2.4
Node 5-A	-1244	-2569	-6107	-70.1	-624.5	41.7
Node 6-A	-2898	1724	-2711	-2565.6	-2199.9	197.7
Node 1-B	-1927	1196	-1555	-989.4	-1154.9	-180.1
Node 1-C	-3084	468	-1261	-906.6	-1250.8	63.5
Node 2-C	2134	298	-1920	-2604.2	-2119.9	479.4
Node 1-D	-2565	1878	-1334	247.7	-193.3	16.6
Node 1-E	-6540	7508	3930	-4915.7	-6267.6	810.2
Node 2-E	-5013	1175	2933	23.4	691.6	-28.4
Node 3-E	-3846	-512	-161	-22.6	-465.8	-51.0
Node 4-E	-5240	-1714	-3136	-134.3	-787.8	147.6
Node 5-E	-6641	130	3600	-23.8	559.3	1.0
Node 6-E	-2557	-56	-670	-30.1	393.8	8.8
Node 7-E	-2714	-66	458	4.9	-982.9	7.4
Node 8-E	-3983	-1261	406	-166.6	-1164.6	173.5
Node 9-E	-6805	17667	1385	362.3	-3583.0	-372.1
Node 10-E	-2694	1296	7474	68.6	-1151.1	-61.4
Node 11-E	-4066	-27	53	-24.2	-497.1	8.2
Node 12-E	-3452	-123	-60	-3.9	-190.3	10.6
Node 13-E	-2872	-88	-64	-0.8	51.5	14.6
Node 14-E	-2434	-48	-35	-0.4	155.8	15.4
Node 15-E	-2412	-500	-799	-2.6	343.3	-4.7
Node 16-E	-2201	-1069	3704	-826.0	-1358.1	67.1
Node 17-E	-1469	-32	601	-0.3	186.3	8.1
Node 18-E	-1161	5	70	0.0	127.0	-2.3
Node 19-E	-1074	66	44	0.3	84.3	-0.8
Node 20-E	-860	406	-599	2.3	116.6	-7.6
Node 21-E	-1090	69	966	-67.7	-1159.9	-17.8
Node 1-F	-5091	1631	-10177	3143.4	-2639.7	-103.1
Node 2-F	-5206	-1139	3414	-25.5	958.8	25.2
Node 3-F	-3806	541	-202	18.9	-465.6	50.9
Node 4-F	-5167	7178	2960	128.3	-1272.6	166.7
Node 5-F	-4572	-121	3400	37.3	469.7	-4.6
Node 6-F	-2539	21	-559	30.9	357.0	-6.4
Node 7-F	-2776	18	438	5.0	-918.9	-6.1
Node 8-F	-4987	-237	-3599	34.5	1652.3	-59.8
Node 9-F	-6465	-17254	965	-314.4	-3347.0	410.5
Node 10-F	-5086	1229	6807	-42.4	-2619.6	45.3
Node 11-F	-3952	75	-53	-0.1	-499.2	-9.8
Node 12-F	-3398	106	-113	2.2	-197.8	-3.9
Node 13-F	-2831	81	-51	0.8	36.9	-15.1
Node 14-F	-2484	24	-12	1.2	167.4	-17.7
Node 15-F	-2523	265	-663	8.7	455.6	12.3
Node 16-F	-3406	371	684	8.7	-2019.9	-36.4
Node 17-F	-2777	-42	218	0.4	251.7	-15.8
Node 18-F	-2538	31	19	0.5	40.9	-12.5
Node 19-F	-2489	-7	47	-0.3	-148.1	-6.1
Node 20-F	-2896	111	-591	2.3	168.8	-1.8
Node 21-F	-2126	273	2655	-603.4	-2162.2	-44.2

1.0 Seismic Load X

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 1-A	-2476	704	-243	7.9	-1802.3	5.7
Node 2-A	-2290	128	186	3.0	232.9	14.3
Node 3-A	-2369	-37	21	-0.9	29.0	12.0
Node 4-A	-2378	-36	66	-2.1	-129.2	5.7
Node 5-A	-2596	-459	-614	-10.0	182.6	-2.3
Node 6-A	-2249	-96	2769	568.1	-1738.8	57.8
Node 1-B	-869	-771	-1904	-114.2	-682.6	-192.6
Node 1-C	-1100	-111	-1258	5.5	-956.0	76.1
Node 2-C	-1172	247	1302	40.7	-959.4	56.5
Node 1-D	-1149	286	-1696	250.7	-1034.0	24.0
Node 1-E	-5530	-1768	-9781	-2747.4	-2161.4	192.0
Node 2-E	-5427	-1256	3077	25.1	734.2	-28.3
Node 3-E	-4187	-551	-178	-24.4	-508.4	-54.6
Node 4-E	-5694	-1733	-3366	-144.2	-846.3	159.1
Node 5-E	-7182	142	3865	-25.5	595.7	1.2
Node 6-E	-2783	-53	-670	-31.3	408.4	10.9
Node 7-E	-2967	-71	465	5.3	-1030.0	7.5
Node 8-E	-7272	-56	-4205	38.0	1689.2	65.8
Node 9-E	-7370	18968	1563	389.2	-3858.6	-399.0
Node 10-E	-2694	1296	7474	68.6	-1151.1	-61.4
Node 11-E	-3955	-31	22	-9.6	332.9	7.4
Node 12-E	-3261	-126	-93	-3.8	-185.6	8.7
Node 13-E	-2627	-76	-43	-0.5	47.3	13.4
Node 14-E	-2229	-58	-263	-0.6	142.8	14.1
Node 15-E	-2232	-401	-395	-2.1	229.1	-2.5
Node 16-E	-2158	-362	2741	-611.5	-1005.0	59.2
Node 17-E	-1536	7	564	0.0	179.1	7.5
Node 18-E	-1175	-20	74	-0.1	121.2	-3.1
Node 19-E	-988	-13	90	0.2	87.5	-2.2
Node 20-E	-1116	244	425	2.2	166.6	-4.2
Node 21-E	-1206	-139	809	-80.5	-1275.5	-21.3
Node 1-F	-5494	1729	-10932	3264.1	-2827.8	-108.0
Node 2-F	-5634	-1216	3591	-27.4	1014.5	27.0
Node 3-F	-4145	582	-222	20.3	-508.1	54.5
Node 4-F	-5593	7736	-3177	137.7	-1328.9	166.7
Node 5-F	-5107	-131	3601	29.3	499.5	-5.0
Node 6-F	-2763	15	-557	32.3	371.4	-8.3
Node 7-F	-3232	19	446	5.6	-968.9	-6.1
Node 8-F	-7534	-252	-3879	-36.8	1762.8	-64.6
Node 9-F	-6996	-18547	1125	-337.5	-3608.2	440.6
Node 10-F	-5482	1323	7337	-45.4	-2822.8	45.3
Node 11-F	-3875	11	16	-4.8	-331.0	-8.1
Node 12-F	-3247	113	-103	3.1	-187.0	-5.9
Node 13-F	-2624	76	-48	-0.6	34.6	-13.2
Node 14-F	-2302	40	-174	0.5	207.1	-15.4
Node 15-F	-2339	188	-427	5.5	331.1	6.3
Node 16-F	-3177	46	275	3.9	1974	-33.8
Node 17-F	-2536	-25	102	-0.5	197.0	-12.4
Node 18-F	-2412	44	14	0.8	54.8	-12.1
Node 19-F	-2489	21	17	0.9	-156.0	-6.5
Node 20-F	-2704	180	-531	5.5	76.4	2.0
Node 21-F	-2216	120	2715	-496.0	-1699.8	-54.0

1.0 Seismic Load Y

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 1-A	-895	404	-4253	60.8	471.6	486.4
Node 2-A	166	626	-1636	84.2	-197.8	17.9
Node 3-A	133	706	-1731	95.6	-252.3	-212.5
Node 4-A	77	1002	-1684	110.7	-112.4	70.3
Node 5-A	-375	2118	-1674	108.8	88.7	186.8
Node 6-A	350	-2691	-3163	2211.8	-458.5	-227.4
Node 1-B	73	-1510	-3157	569.2	103.4	-157.1
Node 1-C	-1023	-4175	-3535	1399.5	351.4	-180.0
Node 2-C	-2013	-5007	-2099	1664.3	-296.5	200.1
Node 1-D	2003	-2957	2099	1376.7	-14.1	-94.6
Node 1-E	-1495	-1491	-13165	1157.0	2527.8	-156.7
Node 2-E	391	-2421	-9073	97.4	-52.4	213.7
Node 3-E	352	-5019	-8410	2291.7	278.9	-80.6
Node 4-E	-693	-7223	-10254	94.1	2710.9	202.8
Node 5-E	-954	-6934	-9093	230.1	275.1	-112.8
Node 6-E	-2	-5663	-9600	1307.4	487.4	-68.5
Node 7-E	-956	-5184	-11710	2325.4	-892.1	164.4
Node 8-E	-2722	-56	-4205	38.0	1689.2	65.8
Node 9-E	-7370	18968	1563	389.2	-3858.6	-399.0
Node 10-E	-2694	1296	7474	68.6	-1151.1	-61.4
Node 11-E	-3955	-31	22	-9.6	332.9	7.4
Node 12-E	-751	-569	-1827	105.3	43.9	-473.3
Node 13-E	-2627	-76	-43	-0.5	47.3	-370.3
Node 14-E	-2229	-58	-263	-0.6	142.8	-147.5
Node 15-E	-780	-892	1808	26.7	164.0	126.6
Node 16-E	-948	-12205	1583	766.7	-275.4	-202.4
Node 17-E	-89	-4760	-637	-2.8	-101.0	-303.4
Node 18-E	25	-5639	-242	5.4	9.9	-26.9
Node 19-E	40	-6015	-1008	5.3	-19.1	45.0
Node 20-E	164	-7926	-300	-10.2	-19.7	88.4
Node 21-E	96	-6231	-22	-996.5	-1.3	57.0
Node 1-F	-1101	-7378	13175	2525.0	-2551.4	-217.8
Node 2-F	-329	-2361	9639	98.7	63.4	211.0
Node 3-F	-166	-5055	8473	2290.6	-299.8	-82.6
Node 4-F	-740	-7274	10184	92.8	-2654.8	202.4
Node 5-F	-753	-4974	9165	2300.7	292.9	-116.1
Node 6-F	-217	-5685	9709	1306.3	-476.8	-86.2
Node 7-F	-1207	-5176	11624	2329.4	605.2	159.6
Node 8-F	-419	-2967	9499	113.2	165.8	-271.7
Node 9-F	-172	-10469	14820	2375.4	822.6	416.5
Node 10-F	-952	-10952	13175	2525.0	-2551.4	-217.8
Node 11-F	-1101	-7378	13175	2525.0	-2551.4	-217.8
Node 12-F	62	-524	1828	104.5	-70.6	-472.5
Node 13-F	-894	-985	1636	87.4	-23.3	375.4
Node 14-F	-1760	-561	1845	76.9	-59.3	-70.2
Node 15-F	-500	172	1932	73.9	-60.4	71.9
Node 16-F	-1133	802	1962	89.7	-30.7	-65.3
Node 17-F	-464	145	1950	75.4	150.8	-101.4
Node 18-F	37	530	2043	84.6	292.6	-217.7
Node 19-F	-133	802	1962	89.7	99.3	78.4
Node 20-F	-1704	180	-531	5.5	76.4	2.0
Node 21-F	-333	-2176	3205	1972.5	488.6	-254.7

REVISION	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A			A01	20/03/2020	ISSUED FOR APPROVAL	--	--	--	--	

SIEMENS
Ingenuity for life

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة المعادن العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

DRAWING NUMBER			REVISION
A1-MDN-C1907-E-1022-001			A
DATE	20/03/2020	DATE 20/03/2020	DATE 20/03/2020
DRAWN	EDC	CHECKED AC	APPROVED VS

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_950_E_V_29504_001	0
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707__GAU01__WD01_EQS03__45013	OU100676836 / 0

DRAWN	--	--
CHECKED		
ENGINEER	--	--
APPROVED		
EMPLOYER		
SCALE	1:XX	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
POWER DISTRIBUTION
ELECTRICAL EQP LAYOUT PREF SUBSTATION # 01

DRAWING NUMBER	REV
MD-216-49PQ-EG-EL-LAD-V552-1001	A02